

Zhaodong Liu

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EDUCATION

New York University Shanghai

Shanghai, China

Bachelor of Science; Double Major: Computer Science, Mathematics

Sep 2022-May 2026 (expected)

Cumulative GPA: **3.811/4.0** | CS Major GPA: **3.941/4.0** | Math Major GPA: **3.926/4.0**

- **Study-away Experience:** NYU New York, Fall 2024 and Spring 2025
- **Relevant Coursework:** Multivariable Calculus, Honors Linear Algebra, Probability & Statistics, Honors Numerical Analysis, Discrete Mathematics, Analysis, Introduction to Computer Science & Data Science, Data Structures, Basic Algorithms, Introduction to Databases, Natural Language Processing, Java and Web Design
- **Honors:** Dean's List (Academic Year 2022-2023, 2023-2024, 2024-2025)

RESEARCH EXPERIENCE

Research Assistant, New York University Shanghai

May 2025-present

Generative Retrieval for Recommendation System

Advisor: Prof. Hongyi Wen, Department of Computer Science, Data Science and Engineering

- Developed models to predict user behavioral patterns on Amazon datasets.
- Designed and evaluated model architectures, including RQ-VAE, VQ-VAE, and OPQ, to capture discrete behavior patterns, achieving 25.6% performance improvement (from 0.0359 to 0.0451).
- Implemented advanced tokenization methods, including BPE-inspired partitioning, improving sequence coherence and representation quality.
- Adapted the model framework to support multimodal representation learning.

Research Assistant, New York University

Sep 2024-Dec 2024

Finetuning on Pretrained LM to Efficiently Perform Retrosynthesis of Chemical Reactions

Advisor: Prof. He He, Center for Data Science

- Developed predictive models for chemical reaction retrosynthesis using SMILES notation.
- Applied an LLM for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (from 0.588 to 0.742) while reducing reliance on expensive laboratory data.
- Employed fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to optimize training efficiency.

Research Assistant, New York University Shanghai

May 2024-Aug 2024

Efficient Driving Route Optimization Considering Traffic Light Timings

Advisor: Prof. Zhibin Chen, Department of Computer Science, Data Science and Engineering

- Built a traffic-aware route optimization system under real-world constraints.
- Applied Dijkstra's algorithm and related methods to compute the optimal path(s).
- Simulated model performance on real traffic signal datasets, demonstrating significant reduction in travel time compared to baseline routing algorithms, while maintaining efficiency.

LEADERSHIP & ACTIVITIES

PEER-Pal Club, New York University Shanghai| Treasurer

May 2023-Aug 2024

- Managed financial operations and annual budgeting for student-led educational initiatives.
- Organized exhibitions and club fairs, strengthening engagement with the campus community.
- Facilitated letter exchanges and Q&A sessions with high school students from Hunan Province.

Chamber Orchestra, New York University Shanghai| Trombone Section Leader

Sep 2022-present

- Led the trombone section, coordinated rehearsals, and oversaw instrument storage logistics.

DISTINCTIONS

Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention

May 2023

- Preprocessed large datasets to remove anomalies and ensure data quality.
- Built predictive models using methods including Bayesian regression for market analysis.
- Synthesized regional market data to support modeling results.

SKILLS AND OTHERS

- **Programming:** Python, Java, C, SQL, Stata
- **Frameworks & Tools:** Git, PyTorch, Pandas, NumPy, LaTeX
- **Languages:** Chinese (Native), English (Fluent)